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$$a) A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 2 \\ -1 & 3 & 1 \end{bmatrix}$$

$$(A|I) \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ -1 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_1 \leftarrow R_1 + R_2 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 5 & 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ -1 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_1 \leftarrow R_1 - 5R_2 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 1/5 & 0 & 1/5 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ -1 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_2 \leftarrow R_2 - R_1 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 1/5 & 0 & 1/5 \\ 0 & 0 & 2 & -1/5 & 1 & -1/5 \\ -1 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_2 \leftarrow R_2 \cdot 1/2 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 1/5 & 0 & 1/5 \\ 0 & 0 & 1 & -1/10 & 1/2 & -1/10 \\ -1 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_1 \leftarrow R_1 + 1/2 R_2 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 1/5 & 0 & 1/5 \\ 0 & 0 & 1 & -1/10 & 1/2 & -1/10 \\ 1 & 0 & 0 & 1/2 & 1/2 & -1/2 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 1/2 & -1/2 \\ 0 & 1 & 0 & 1/5 & 0 & 1/5 \\ 0 & 0 & 1 & -1/10 & 1/2 & -1/10 \end{array} \right]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 1/2 & 1/2 & -1/2 \\ 1/5 & 0 & 1/5 \\ -1/10 & 1/2 & -1/10 \end{bmatrix}$$

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b)

$$\Delta = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & -1 & 2 \\ 0 & 0 & 5 & 3 \\ 0 & 0 & -2 & 1 \end{vmatrix}$$

$$\Delta = 1 \cdot \begin{vmatrix} 1 & -1 & 2 \\ 0 & 5 & 3 \\ 0 & -2 & 1 \end{vmatrix} - 2 \cdot \begin{vmatrix} 2 & -1 & 2 \\ 0 & 5 & 3 \\ 0 & -2 & 1 \end{vmatrix}$$

$$+ 3 \cdot \begin{vmatrix} 2 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 1 \end{vmatrix} - 4 \cdot \begin{vmatrix} 2 & 1 & -1 \\ 0 & 0 & 5 \\ 0 & 0 & -2 \end{vmatrix}$$

$$= 1[5 + 0 + 0 - 0 + 6 - 0]$$

$$- 2[10 + 0 + 0 + 12 - 0]$$

$$+ 3[0 + 0 + 0 - 0 - 0 - 0]$$

$$- 4[0 + 0 + 6 - 0 - 0 - 0]$$

$$= 11 - 44 = -33$$